

Code No: V3121

R07

Set No: 1

III B.Tech. I Semester Supplementary Examinations, December - 2013

DIGITAL IC APPLICATIONS

(Electronics & Communication Engineering)

Time: 3 Hours

Max Marks: 80

Answer any FIVE Questions
All Questions carry equal marks

1. a) Design CMOS transistor circuit for 2-input AND gate. Explain the circuit with the help of function table.
b) Draw the resistive model of a CMOS inverter circuit and explain its behaviour for LOW and HIGH outputs.
2. a) Explain the terms (i) DC noise margin (ii) Fan-out with reference to TTL gate.
b) Draw the circuit diagram of two-input 10K ECL OR gate and explain its operation.
3. a) Write a VHDL Entity and Architecture for a 3-bit synchronous counter using Flip-Flops.
b) Explain the use of Packages. Give the syntax and structure of a package in VHDL.
4. a) Design a combinational circuit for the function $f(a,b,c) = \pi(1,2,4,6,7)$
b) Write the VHDL program for the above function by structural model.
5. a) What is the function of 74XX138 and prove it?
b) Design a function $F = ABC + (A+B+C)'$ by using 74XX138.
6. What is a comparator? Explain the operation of a 2-bit comparator with a relevant diagram. Draw its logic symbol and write a VHDL code for it.
7. a) Design an 8-bit universal parallel-in serial out shift register with a control input? Shift-left operation with control input 1 and shift-right operation with control input 0 is to be performed?
b) Design a serial binary adder? Develop the VHDL program for simulating serial binary adder?
8. a) Explain the necessity of two-dimensional decoding mechanism in memories. Draw MOS transistor memory cell in ROM and explain the operation.
b) Explain the internal structure of 64K X 1 DRAM with the help of timing waveforms.

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Answer any FIVE Questions
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1. a) Explain how to estimate sinking current for low output and sourcing current for high output of CMOS gate.
b) List out the family members of TTL family.
2. a) Draw the circuit diagram of basic CMOS gate and explain the operation.
b) Differentiate CMOS and TTL gates.
3. a) Discuss different delays and concept of packages in VHDL.
b) Explain about design flow of VHDL.
4. a) Explain structural design elements of VHDL.
b) Design the logic circuit and write a data-flow style VHDL program for the following function $F(R) = f(A,B,C,D) = \pi(1, 4, 5, 7, 9, 13, 15)$.
5. Design a 16-bit ALU using 74XX381 and 74XX182 ICs.
6. a) Design a 16-bit comparator using 74 XX85 ICs.
b) Write the VHDL program for the above IC.
7. Explain the operation of a 4 bit synchronous binary counter with the required diagram and waveforms.
8. a) With the help of timing waveforms, explain read and write operations of SRAM.
b) Explain the internal structure of 64K X 1 DRAM. With the help of timing waveforms discuss DRAM access.

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DIGITAL IC APPLICATIONS

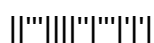
(Electronics & Communication Engineering)

Time: 3 Hours

Max Marks: 80

Answer any FIVE Questions
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1. a) Explain the terms Logic levels and Noise margin of CMOS logic.
b) Analyze the fall time CMOS inverter output with $R_L = 100$, $V_L = 2.5V$ and $C_L = 10pF$. Assume V_L as stable state voltage.
2. a) What is the necessity of separate interfacing circuit to connect CMOS gate to TTL gate?
b) How the CMOS family is advantage over TTL family?
3. a) Write a VHDL Entity and Architecture for the function $F(x) = a + b + c$. Also draw the relevant logic diagram.
b) Explain the use of Packages Give the syntax and structure of a package in VHDL.
4. a) Design a logic circuit to detect prime number of a 4-bit input? Write the VHDL program for the same in structural style of modelling.
b) Design the logic circuit and write a data-flow style VHDL program for the Function $F(X) = \sum A, B, C, D(3, 5, 6, 7, 10, 13, 14) + d(1, 2, 4, 15)$.
5. Explain clearly about 74 X 148 priority encoder with its logic symbol. Give its truth table and write VHDL code in any one of the model.
6. a) Design a 16-bit comparator using 74X85 ICs.
b) Write a behavioural VHDL program to compare 16-bit signed and unsigned integers.
7. a) Write VHDL code for 4-bit serial-In Parallel-out register.
b) Explain the difference between Latch and Flip Flop.
8. a) Explain the operation of Synchronous SRAM with the help of its internal architecture.
b) With a neat diagram explain the general architecture of CPLD? Discuss the key features of Xilinx XC9500 CPLD family.



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Answer any FIVE Questions
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1. a) Design a CMOS 2 I/P NOR gate and verify the functional verification.
b) Draw the transfer characteristics of CMOS inverter and explain.
2. a) Explain sinking current and sourcing current of TTL gate. Which of the parameters decide the fan-out and how?
b) Expand HC, HCT, and VHC,& VHCT in connection with CMOS family.
3. a) Discuss the steps in VHDL design flow.
b) Explain the difference in program structure of VHDL and any other procedural language. Give an example.
4. Design the logic circuit and write a data-flow style VHDL program for the following functions?
a) $F(A) = \pi p,q,r,s(1,2,3,4,5,6,7,12,13,14)$
b) $F(X) = \sum A,B,C,D(1,2,3,5,6,7,13) + d(1,2,4,12,15)$
5. Design a 16-bit ALU using 74x381 and 74x182 ICs?
6. a) Draw the logic diagram of 74x74 IC and explain the operation? Develop the VHDL model for this IC?
b) With the help of logic diagram discuss PAL16R8?
7. Explain the operation of a 4 bit synchronous binary counter with the required diagram and wave forms.
8. Write short notes on the following:
a) RAM b) SSRAM c) DRAM d) SDRAM.
